

SOLVED PRACTICE WORKBOOK

Geography 22 - India Himalayan Temperate Forests - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

Himalayan Temperate Forests

Altitude × monsoon × winter snow × aspect × disturbance × institutions



Use the mosaic as an answer spine: locate → explain mechanism → qualify → govern → monitor.

Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge → official current linkage → clearly marked analysis. No fabricated PYQs or statistics.

Part II - Premium solved practice workbook

Workbook rules

- Objective answers rotate continuously **A -> B -> C -> D** from Q1 through Q32.
- Authentic PYQ option order is retained; PYQs are placed at rotation-compatible positions.
- “Officially verified” is used only where a locally available official key exists.
- The relevant 2019, 2020 and 2022 objective keys were not held locally; answers are retained as **INFERRED ANSWER — NOT OFFICIALLY VERIFIED**, with confidence and elimination.
- Every Mains solution follows **claim -> named evidence/example -> analysis -> qualification -> verdict** and ends **Why this earns marks**.

A. Solved Prelims PYQs, hard MCQs and remedial MCQs

Q1. UPSC Prelims 2020, GS-I, Q77 — official key unavailable locally

Which of the following are the most likely places to find the musk deer in its natural habitat?

1. Askot Wildlife Sanctuary
2. Gangotri National Park
3. Kishanpur Wildlife Sanctuary
4. Manas National Park

A. 1 and 2 only B. 2 and 3 only C. 3 and 4 only D. 1 and 4 only

INFERRED ANSWER — NOT OFFICIALLY VERIFIED: A. Confidence: high.

Elimination: Askot and Gangotri are high Himalayan protected landscapes suitable for musk-deer habitat. Kishanpur is a Terai protected area and Manas is a foothill-alluvial/subtropical landscape rather than the paired high-Himalayan habitat tested. The answer is not upgraded to official status because the local official key was unavailable.

Q2. Original — classification firewall

Which statement is most defensible?

A. Very Dense Forest is an old-growth ecological class. B. Forest type, FSI canopy class, recorded forest area and legal category are separate descriptors. C. Every recorded forest is densely wooded. D. A plantation cannot enter forest-cover statistics.

Answer: B.

Explanation: FSI density, ecological formation, government record and legal status answer different questions. A plantation may qualify as forest cover under the methodology.

Q3. UPSC Prelims 2022, GS-I, Q45 — stem verified; official key unavailable locally

With reference to “Gucchi” sometimes mentioned in the news, consider the following statements:

1. It is a fungus.
2. It grows in some Himalayan forest areas.
3. It is commercially cultivated in the Himalayan foothills of north-eastern India.

Which of the statements given above is/are correct?

A. 1 only B. 3 only C. 1 and 2 D. 2 and 3

INFERRED ANSWER — NOT OFFICIALLY VERIFIED: C. Confidence: high.

Elimination: Guichi refers to morel mushrooms (*Morchella* spp.), a fungus collected in Himalayan forest landscapes. The statement of established commercial cultivation in the north-eastern Himalayan foothills is not supported. The exact answer remains inferred because an official local key was unavailable.

Q4. UPSC Prelims 2019, GS-I, Q18 — official scan verified; key unavailable locally

Which one of the following National Parks lies completely in the temperate alpine zone?

A. Manas National Park B. Namdapha National Park C. Neora Valley National Park D. Valley of Flowers National Park

INFERRED ANSWER — NOT OFFICIALLY VERIFIED: D. Confidence: high.

Elimination: Manas is a foothill/alluvial landscape. Namdapha spans a very large tropical-to-alpine gradient. Neora Valley also spans montane belts. Valley of Flowers is the high Himalayan option identified with the temperate-alpine setting. “Temperate alpine” is a PYQ phrase; ecological teaching should still distinguish upper forest, subalpine scrub and alpine meadow.

Q5. Original — physical template

Which sequence best explains the actual forest patch on a Himalayan slope?

A. Regional circulation -> range exposure -> altitude/aspect/topoclimate -> soil/disturbance -> vegetation outcome B. State boundary -> latitude only -> tree species C. Elevation alone -> fixed belt everywhere D. Canopy density -> legal status -> climate

Answer: A.

Explanation: The Himalayan forest mosaic is nested. Regional circulation and range position set moisture, then local topoclimate, soil and disturbance filter species and regeneration.

Q6. Original — western/eastern contrast

Which comparison is correct?

A. Eastern temperate forests are merely western species receiving more rain. B. Eastern forests combine stronger monsoon/cloud influence and distinct biogeographic links; western forests show stronger inner-valley dry-temperate and winter-snow contrasts. C. Western disturbances uniformly control eastern Himalayan summer rain. D. Dry temperate forest is absent from all Himalayan rain shadows.

Answer: B.

Explanation: The contrast is climatic and biogeographic. Local exceptions remain in both sectors.

Q7. Original — chir pine

Which statement about chir pine is correct?

A. It is the same species as blue pine/kail. B. It is an upper subalpine fir. C. It is mainly a subtropical/lower mountain pine and must not represent all Himalayan temperate conifers. D. It cannot carry surface fire.

Answer: C.

Explanation: Chir is *Pinus roxburghii*. Blue pine is *Pinus wallichiana*. Fir belongs to *Abies*.

Q8. Original — upper conifer distinctions

Which pair is incorrectly treated as interchangeable in a careless answer?

A. Deodar and cedar B. Blue pine and kail C. Spruce and *Picea smithiana* D. Fir and spruce as one species

Answer: D.

Explanation: Fir and spruce are distinct genera and ecological components. Common-name/scientific-name discipline matters.

Q9. Original — aspect

On a western Himalayan slope, which is the safest default expectation?

A. South-facing slopes often receive more solar energy and may be warmer/drier, while local cloud, snow and relief can modify the pattern. B. North-facing slopes are always drier. C. Aspect has no effect below treeline. D. East-facing slopes are universally wettest.

Answer: A.

Explanation: The Northern Hemisphere radiation rule is useful only with a qualification.

Q10. Original — soils

Why is “every conifer soil is a podzol” incorrect?

A. Podzolisation occurs only in deserts. B. Leaching, litter, parent material, drainage, erosion and time must jointly favour the process. C. Conifers produce no litter. D. Mountain soils never form horizons.

Answer: B.

Explanation: Podzolisation is a process tendency, not a canopy label.

Q11. Original — broadleaf/conifer water relations

Which statement is best?

A. Broadleaf forest always produces more streamflow than conifer forest. B. Conifers always reduce soil water. C. Catchment response depends on species, stand age, season, interception, evapotranspiration, soil and snow; leaf form alone is insufficient. D. Forest has no hydrological effect.

Answer: C.

Explanation: Water claims require scale and measurement discipline.

Q12. Original — succession

Which is the most accurate description of Himalayan forest succession?

A. It always ends in one permanent climatic climax. B. Fire always converts forest to grassland permanently. C. A plantation is automatically the climax stage. D. Disturbance history, seed source, browsing, fire and chance can produce multiple recovery pathways.

Answer: D.

Explanation: Modern succession ecology is probabilistic and patch-based.

Q13. Original — oak–pine debate

Which monitoring design best tests alleged pine expansion into oak forest?

A. Compare mapped canopy with seedling/sapling, fire, browse, soil moisture, aspect, seed source and land-use history data. B. Record only adult pine trees in one year. C. Assume every pine patch is caused by climate change. D. Compare state-level rainfall with one photograph.

Answer: A.

Explanation: Regeneration and disturbance data test mechanism; canopy alone cannot.

Q14. Original — fire metrics

Which pair must be kept distinct?

A. Surface fire and litter B. Active-fire detections and independent incidents/burned area C. Slope and aspect D. Fir and snow

Answer: B.

Explanation: One incident can generate multiple detections, while small fires may be missed. Burned area and severity require separate products.

Q15. Original — ecosystem services

Which claim is defensible?

A. Forest prevents every Himalayan landslide. B. Forest has no role in erosion. C. Roots, litter and roughness can reduce shallow erosion/failure risk, but deep-seated failures and extreme events can exceed buffering. D. Afforestation stabilises any road cut without drainage work.

Answer: C.

Explanation: Mechanism plus limit earns marks.

Q16. Original — ISFR data

Which conclusion cannot be drawn from a patch classified as Very Dense Forest?

A. Canopy density meets the relevant threshold. B. The patch can be compared through a stated remote-sensing method. C. It has high detected canopy closure. D. It is necessarily intact old-growth native temperate forest with secure regeneration.

Answer: D.

Explanation: Naturalness, composition, age and regeneration need field/type evidence.

Q17. Original — legal stack

Which sequence correctly separates instruments?

A. FRA = rights; forest-conservation law = diversion approval; working plan = management plan; CAMPA = finance; monitoring = outcome evidence B. FRA = satellite canopy class; CAMPA = protected area C. EIA = community title; working plan = court order D. Green India Mission = automatic forest clearance

Answer: A.

Explanation: Right, clearance, plan, finance and outcome are separate.

Q18. Original — community institutions

Which statement is most balanced?

A. Every local use is degrading. B. Van panchayats/CFR institutions can support stewardship, but ecological outcome depends on rules, equity, capacity, pressure and monitoring. C. Community rights remove all conservation duties. D. State departments have no role after rights recognition.

Answer: B.

Explanation: Avoid both exclusionary and romanticised extremes.

Q19. Original — restoration hierarchy

Which is the correct order?

A. Plant first -> assess later -> clear old forest if needed B. Compensate -> divert -> measure canopy only C. Avoid intact loss -> protect refugia/corridors -> reduce pressure -> assist native recovery -> monitor D. Replace every mixed forest with one commercially valuable conifer

Answer: C.

Explanation: Avoidance comes before offset/restoration.

Q20. Original — climate evidence

Which statement is correct?

A. One severe fire establishes a long-term climate trend. B. Any upward seedling proves species migration. C. A projection is an observation. D. A robust claim names site, period, variable and method, then separates trend, mechanism and attribution.

Answer: D.

Explanation: Evidence grammar prevents overclaiming.

Q21. Remedial — species and habitat

Which statement is safest?

A. Red panda is an eastern Himalayan range-and-bamboo-associated temperate/subalpine species, not a western-Himalaya-wide forest indicator. B. Musk deer is restricted to subtropical chir forest. C. Takin occurs only in deodar forest. D. Monal proves a stand is old-growth.

Answer: A.

Explanation: Species use ecotones and seasonal ranges; flagship presence does not describe the whole forest.

Q22. Remedial — prescribed fire

Which statement is correct?

A. Prescribed fire is always prohibited and ecologically useless. B. It is a context-specific, authorised professional tool requiring ecosystem, weather, terrain, smoke and escape-risk assessment. C. Every fir-spruce forest requires annual burning. D. Fire severity equals detection count.

Answer: B.

Explanation: A qualified management statement is required.

Q23. Remedial — current forest-law title

The current title of the amended 1980 central forest-conservation law is:

A. Indian Forest Rights Act, 1980 B. National Forest Conservation Code, 1980 C. Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980 D. Compensatory Afforestation Act, 1980

Answer: C.

Explanation: The 2023 amendment renamed, rather than repealed, the Act.

Q24. Remedial — plantation equivalence

Why can compensatory plantation not quickly recreate an old Himalayan temperate forest?

A. Trees cannot grow in mountains. B. Plantations have no carbon. C. Native species are legally prohibited. D. Old trees, soils, fungi, deadwood, genetic structure, hydrology and food webs require long development and site continuity.

Answer: D.

Explanation: The critique is ecological non-equivalence, not opposition to restoration.

Q25. Advanced — monitoring

Which dashboard is most complete?

A. Patch/connectivity + composition/structure + regeneration/browse + fire severity + soil/water/snow + species + rights/compliance B. Plantation count only C. Forest-cover percentage only D. Tourism receipts only

Answer: A.

Explanation: Forest condition is multidimensional.

Q26. Advanced — Eastern Himalaya

Which factor most clearly prevents treating eastern temperate forest as a wetter western duplicate?

A. Eastern Himalaya has no conifers. B. Distinct biogeographic links and floristic turnover combine with monsoon/cloud and steep relief. C. Western disturbances are absent from Earth. D. Eastern forests occur at sea level only.

Answer: B.

Explanation: Composition and evolutionary geography matter alongside climate.

Q27. Advanced — dry temperate

Which statement is correct?

A. Dry temperate forest is always a burned moist forest. B. It is the same as alpine meadow. C. Inner rain-shadow and continental settings can support distinct chilgoza, juniper, deodar or dry broadleaf assemblages. D. It occurs only in Sikkim.

Answer: C.

Explanation: Dry temperate is an ecological setting, not a degradation label.

Q28. Advanced — IIRS 2026 fire case

Which interpretation is valid?

A. The Pauri burned-area estimate proves equal high-severity forest loss. B. MODIS detections equal independent fires exactly. C. The 2026 season proves a long-term climate trend. D. The assessment separates detections, mapped burned area and severity, with most mapped Pauri area classified low severity.

Answer: D.

Explanation: The correct reading preserves measurement categories and time bounds.

Q29. Advanced — old-growth structure

Which field observation most strongly adds information that canopy density alone cannot provide?

- A. A mixed tree-size distribution with large old trees, cavities, standing deadwood, fallen logs and continuous native regeneration
B. The district's total geographical area
C. The colour assigned to the state on a political map
D. Annual tourist arrivals without site location

Answer: A.

Explanation: Old-growth/structural condition requires stand-level evidence. Canopy closure alone cannot reveal age structure, deadwood or regeneration.

Q30. Advanced — snow and hydrology

Which statement about forest and snowmelt is most defensible?

- A. Every conifer canopy always increases annual river flow.
B. Canopy interception, wind redistribution, shading, sublimation, soil storage and storm type jointly determine the seasonal response.
C. Forest has no influence on snow.
D. Tree line and permanent snowline are identical.

Answer: B.

Explanation: Forest-snow hydrology is process- and season-specific; universal yield claims are unsafe.

Q31. UPSC Prelims 2019, GS-I, Q33 — official scan verified; key unavailable locally

Consider the following States:

1. Chhattisgarh
2. Madhya Pradesh
3. Maharashtra
4. Odisha

With reference to the States mentioned above, in terms of percentage of forest cover to the total area of State, which one of the following is the correct ascending order?

- A. 2–3–1–4 B. 2–3–4–1 C. 3–2–4–1 D. 3–2–1–4

INFERRED ANSWER — NOT OFFICIALLY VERIFIED: C. Confidence: high for the report period tested.

Elimination: Maharashtra had the lowest percentage among the four, followed by Madhya Pradesh, Odisha and Chhattisgarh. The question is report-vintage sensitive: never transfer its ranking to a later ISFR without checking the later state table.

Q32. UPSC Prelims 2019, GS-I, Q31 — official scan verified; key unavailable locally

There was growing awareness in India about the importance of Himalayan nettle (*Girardinia diversifolia*) because it is found to be a sustainable source of:

- A. Anti-malarial drug
B. Biodiesel
C. Pulp for paper industry
D. Textile fibre

INFERRED ANSWER — NOT OFFICIALLY VERIFIED: D. Confidence: high.

Elimination: Himalayan nettle yields bast fibre used for textiles and illustrates a non-timber forest/livelihood resource. The item does not establish that every nettle harvest is sustainable; tenure, harvest method and regeneration still matter.

B. Solved Mains PYQs

PYQ 1 — UPSC Mains 2019, GS-I, Q15

Question: How can the mountain ecosystem be restored from the negative impact of development initiatives and tourism? (15 marks, 250 words)

Demand: “How can” requires an implementable restoration strategy, not only a threat list. “Mountain ecosystem” requires slope, forest, water, biodiversity and community integration.

Model solution

- **Claim:** Mountain restoration must begin by preventing new irreversible damage because steep relief, slow soil formation and fragmented altitudinal habitats make post-impact repair costly and incomplete.
- **Named evidence/example:** In the Himalaya, road cutting, hydropower works, pilgrimage/tourism growth and urban expansion can remove forest, alter drainage and destabilise debris. The upper Alaknanda-Kedarnath landscape illustrates the overlap of extreme hydro-geomorphic processes with corridor and tourism exposure; the 2026 IIRS Uttarakhand fire assessment adds a current lower/middle-slope fire-management context.
- **Analysis:** First, map carrying capacity and no-go zones around old forest, springs, avalanche paths, unstable slopes and wildlife corridors. Second, require cumulative basin/corridor appraisal rather than project-by-project clearance. Third, enforce hill-road geometry, drainage, retaining/bioengineering, spoil disposal and post-construction slope audits. Fourth, regulate tourist numbers, construction, waste, water use, traffic and ignition during high-risk periods. Fifth, restore with assisted natural regeneration, native mixed planting, riparian/springshed repair and invasive control. Sixth, recognise van panchayat/FRA institutions and finance community patrol, nursery and monitoring work. Finally, track patch connectivity, regeneration, fire severity, soil erosion, spring discharge and livelihood outcomes.
- **Qualification:** Forests can reduce shallow erosion and runoff but cannot prevent every deep landslide, avalanche or extreme flood. Compensatory plantation cannot rapidly recreate old-growth soils, fungi and structure.
- **Verdict:** Mountain restoration is an **avoid–regulate–repair–co-govern–monitor** programme, not a tree-planting event.

Why this earns marks: It follows the restorative directive, uses named Himalayan evidence, links engineering with ecology and rights, qualifies forest protection claims, and ends with measurable implementation.

PYQ 2 — UPSC Mains 2020, GS-I, Q17

Question: Examine the status of forest resources of India and its resultant impact on climate change. (15 marks, 250 words)

Demand: “Examine” requires evidence, interpretation, counterpoint and a balanced verdict. The answer must separate cover quantity from forest quality.

Model solution

- **Claim:** India has a substantial and monitored forest resource, but canopy extent alone gives an incomplete account of native composition, fragmentation and climate function.
- **Named evidence/example:** ISFR 2023 reports forest cover of **715,342.61 sq km (21.76%)** and tree cover of **112,014.34 sq km (3.41%)**, together **25.17%** of geographical area. Its forest-cover definition is land at least one hectare with canopy density at least 10%, irrespective of ownership/legal status, so plantations and orchards may enter the category.
- **Analysis:** Forests mitigate climate change through biomass, soil and deadwood carbon and through microclimate/water regulation. Himalayan temperate forests add large elevational carbon and biodiversity mosaics. Yet fragmentation, repeated fire, roads, extraction, pests and regeneration failure can reduce resilience even when canopy remains. Fire and degradation release carbon; warming, drought and snow/rain shifts can increase stress. Conversely, rights-based protection, avoided deforestation, assisted natural regeneration and native mixed restoration can improve mitigation and adaptation together.
- **Qualification:** Forest area, tree cover, carbon stock and ecological integrity are not synonyms. A plantation may sequester carbon but cannot quickly replace old-growth habitat and soil. One fire season cannot establish a

climate trend.

- **Verdict:** India's forest-climate policy should retain national canopy monitoring but judge success by native composition, connectivity, regeneration, soil carbon, fire severity, hydrology and equitable governance.

Why this earns marks: It uses a dated official baseline, exposes the metric's blind spot, explains two-way forest–climate interaction, recognises plantation value without claiming equivalence, and proposes outcome indicators.

C. Original solved Mains practice

Original 10-marker — zonation

Question: Explain why altitude alone cannot define the distribution of Himalayan temperate forests. (10 marks, 150 words)

Model solution

- **Claim:** Altitude supplies the broad thermal gradient, but actual forest distribution is a multi-control topoclimatic outcome.
- **Named evidence/example:** The Western Himalaya contains chir transition, oak, deodar, kail, fir-spruce and inner dry-temperate assemblages, while the Eastern Himalaya supports humid oak-laurel-maple, mixed conifer and bamboo-rhododendron transitions. The same indicative montane belt therefore differs longitudinally.
- **Analysis:** Latitude shifts thermal limits; monsoon exposure and rain shadow alter moisture; western disturbances change winter snow; aspect changes radiation and melt; valleys create inversions; ridges face wind; soils, fire, browsing and land-use history filter regeneration.
- **Qualification:** Textbook elevation ranges remain useful as indicative envelopes, not universal contour boundaries.
- **Verdict:** Map relative lower-middle-upper belts, then explain the sector, aspect, moisture and disturbance controls that move them.

Why this earns marks: It answers “why,” contrasts west and east, names physical and ecological filters, and preserves the value—but limits—of altitude.

Original 15-marker — oak–pine debate

Question: “Chir pine is invading Himalayan oak forests because of climate change.” Critically examine. (15 marks, 250 words)

Model solution

- **Claim:** The statement may describe some observed transitions but is too linear and climate-deterministic for the Himalayan mosaic.
- **Named evidence/example:** Chir pine is mainly a subtropical/lower mountain species, while banj, moru and kharsu oaks occupy different temperate niches. Western Himalayan slopes differ by aspect, moisture, repeated fire, grazing, lopping, former cultivation and seed-source history.
- **Analysis:** Fire can kill oak recruits and create open, light-rich sites favourable to pine colonisation. Browsing and lopping can suppress oak regeneration. Warmer/drier conditions may intensify moisture stress and fire weather. Conversely, moist north-facing slopes, fire exclusion, seed-source protection and reduced browse can permit oak recovery; some landscapes remain stable mosaics.
- **Qualification:** Adult canopy snapshots cannot identify causation. Evidence needs seedling/sapling cohorts, fire frequency/severity, browse intensity, soil moisture, aspect, land-use history and multi-year climate data. Chir is not equivalent to all temperate conifers.
- **Verdict:** Frame oak–pine change as **disturbance × moisture × history × climate**, and manage the limiting process rather than demonising one species.

Why this earns marks: It deconstructs the slogan, gives mechanisms on both sides, specifies a testable monitoring design and reaches a nuanced management verdict.

Original 20-marker — integrated governance

Question: Design a climate-resilient conservation and restoration strategy for India's Himalayan temperate forests.
(20 marks, 300 words)

Model solution

- **Claim:** Climate resilience requires protection of spatial heterogeneity and ecological processes, not uniform plantations or upward-shift assumptions.
- **Named evidence/example:** Western deodar-kail-fir-spruce and inner dry-temperate systems contrast with eastern oak-laurel-maple, fir/hemlock and bamboo-rhododendron mosaics. ISFR 2023 provides a canopy baseline but not natural condition. The IIRS 2026 Uttarakhand assessment demonstrates the need to distinguish detections, burned area and severity.
- **Analysis:**
 1. **Avoid:** exclude old-growth, moist refugia, riparian zones, avalanche corridors and unstable slopes from avoidable fragmentation.
 2. **Connect:** protect lateral valley and vertical elevational corridors, including forest–subalpine ecotones.
 3. **Reduce pressure:** fire-weather planning, fuel treatment near assets, grazing/browsing agreements, sustainable logging/NTFP rules, invasive/pest surveillance and tourism regulation.
 4. **Restore:** assisted natural regeneration first; native mixed planting only after soil, drainage and seed-source diagnosis; repair road drainage, gullies and spoil.
 5. **Govern:** converge working plans, FRA/CFR/van panchayats, wildlife corridors, EIA/ESZ conditions, CAMPA and GIM while publishing benefit and compliance data.
 6. **Monitor:** patch/core-edge, composition, regeneration, deadwood, fire severity, soil moisture, snow/hydrology, indicator species, rights and livelihoods.
- **Qualification:** Climate projections guide risk screening but site interventions require local trend and ecological evidence. Plantations can add trees/carbon yet cannot quickly recreate old growth.
- **Verdict:** Resilience is a monitored **refugia–connectivity–diversity–rights** strategy that preserves options under uncertainty.

Why this earns marks: It is regionally differentiated, uses current official evidence carefully, integrates ecology with law and livelihoods, provides an implementation hierarchy and names measurable outcomes.